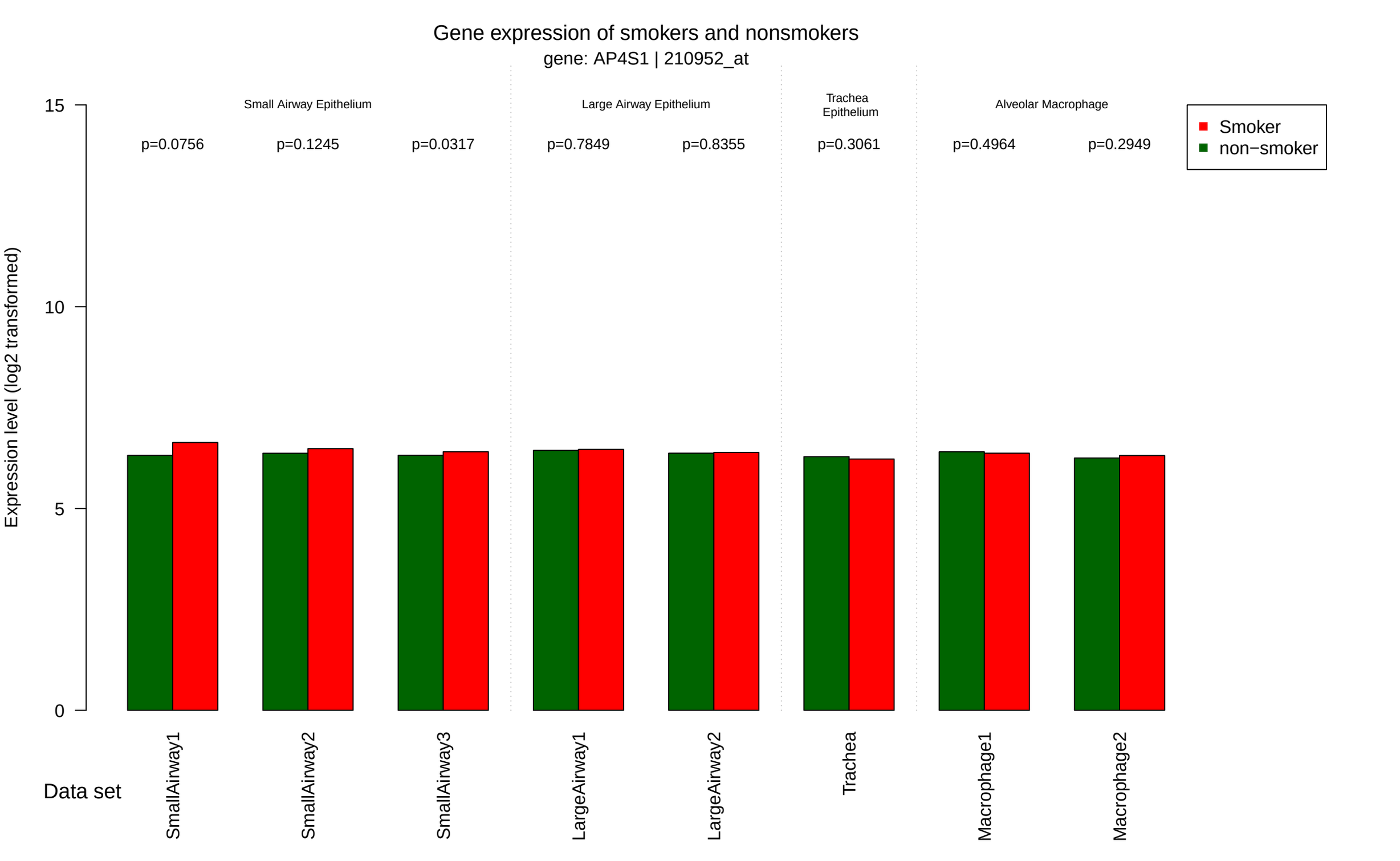
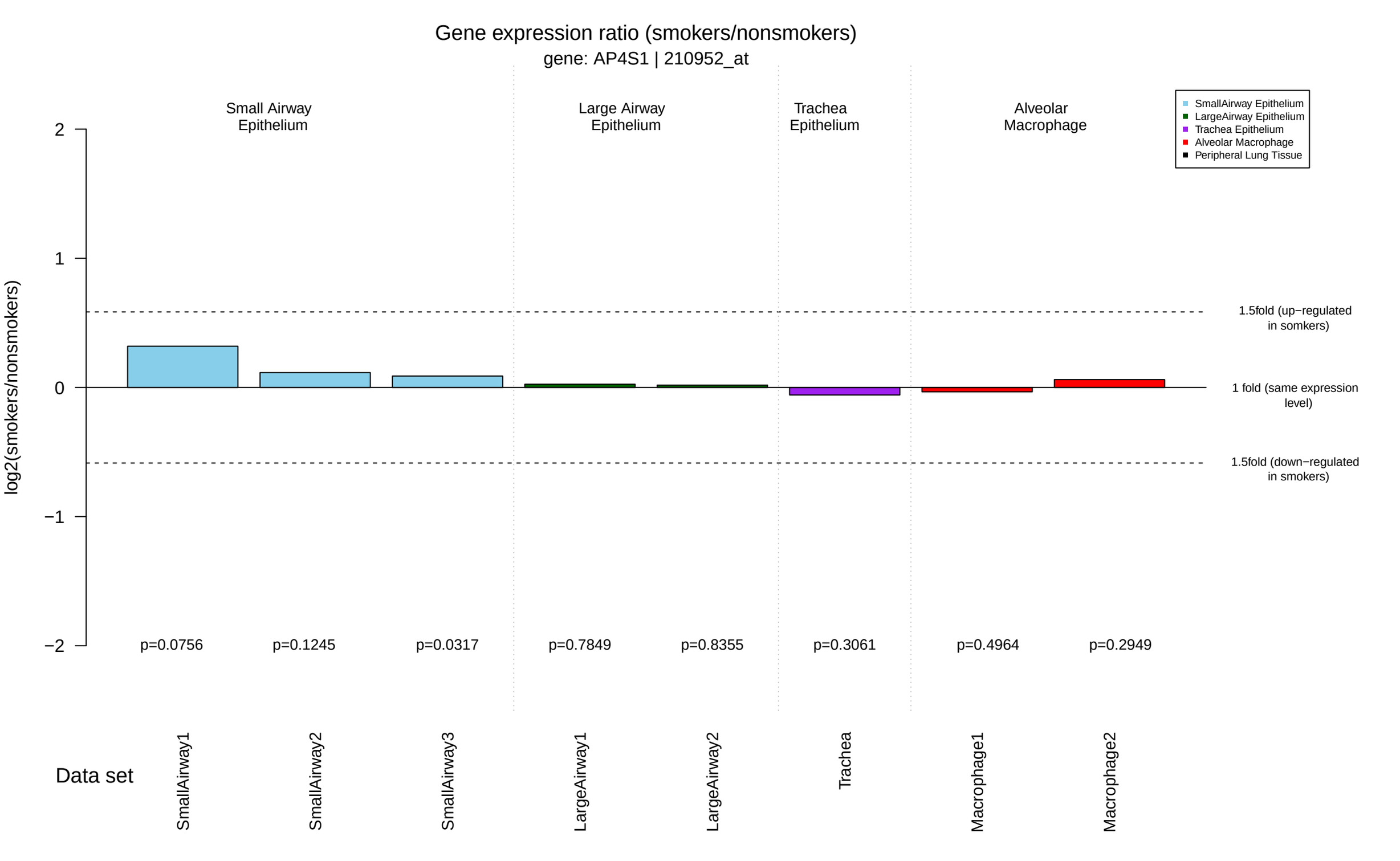
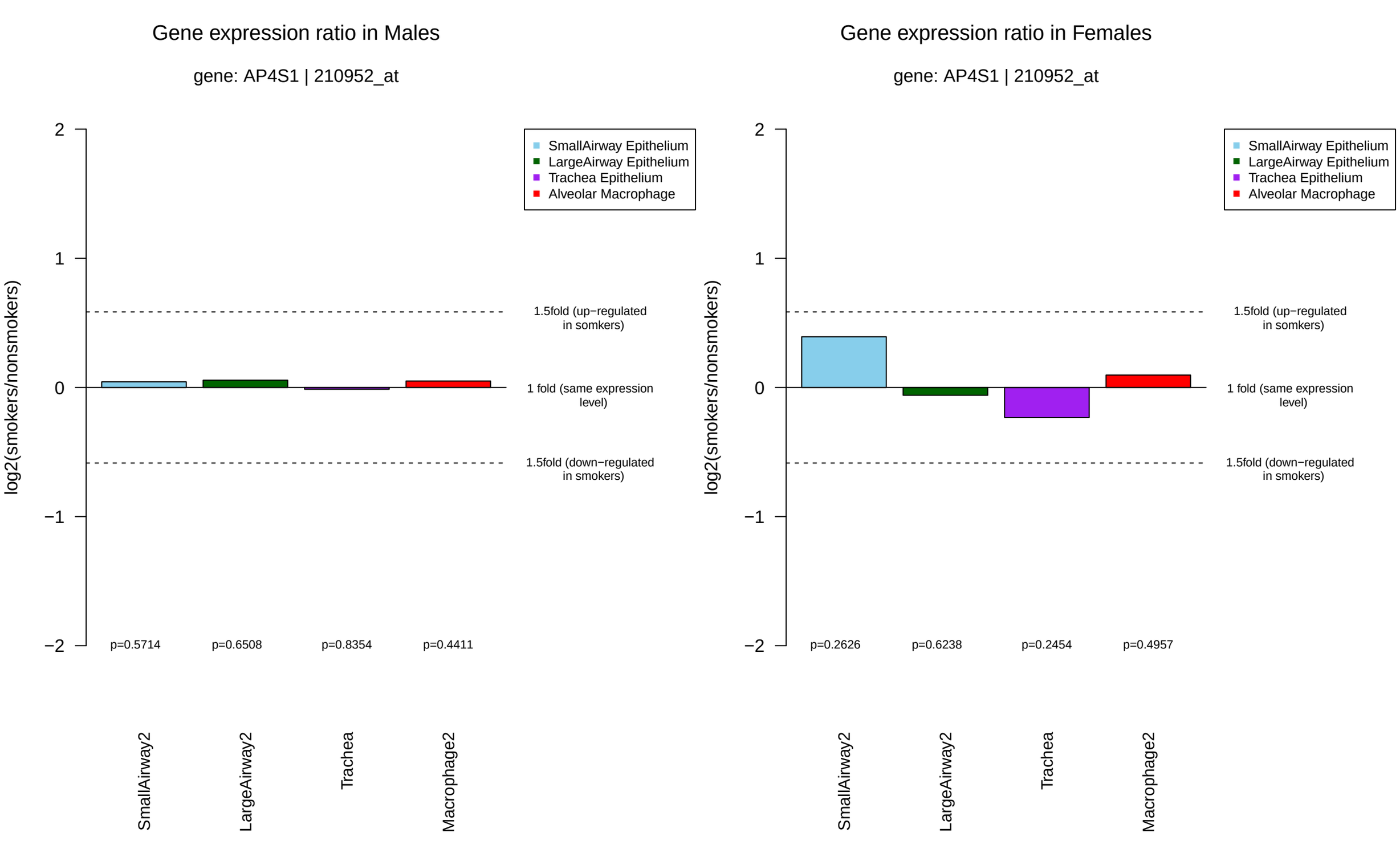
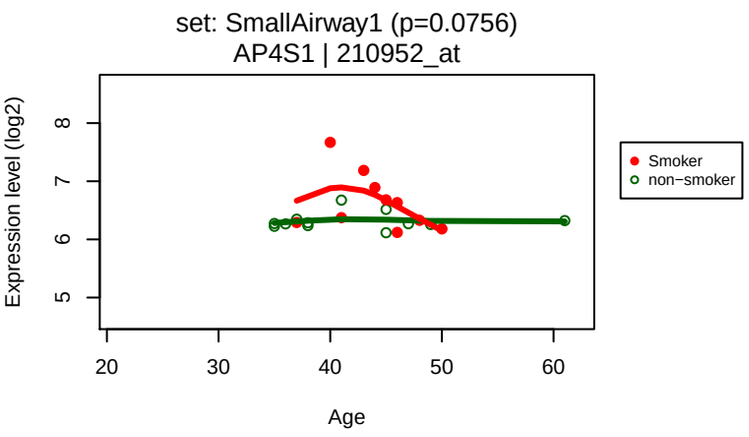
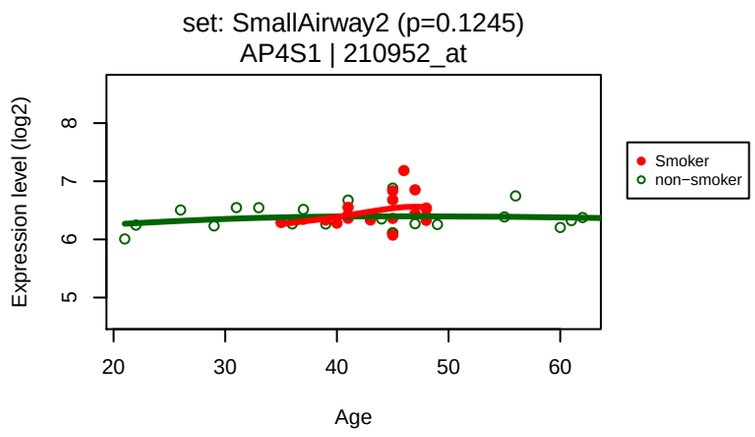
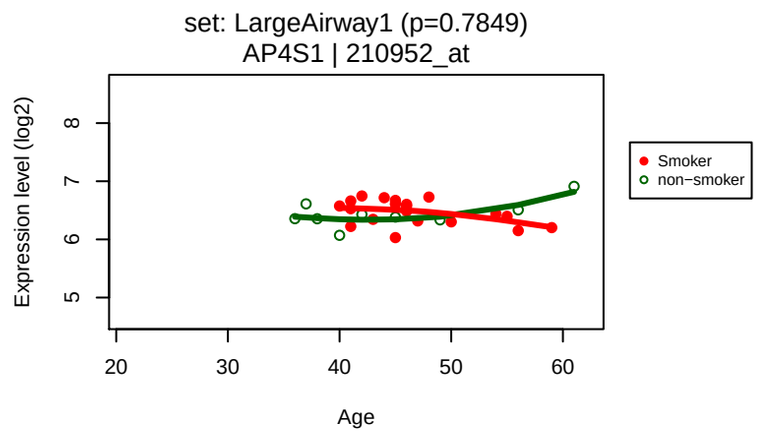

Gene expression of smokers and nonsmokers
gene: AP4S1 | 210952_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0756
p=0.1245
p=0.0317
p=0.7849
p=0.8355
p=0.3061
p=0.4964
p=0.2949
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

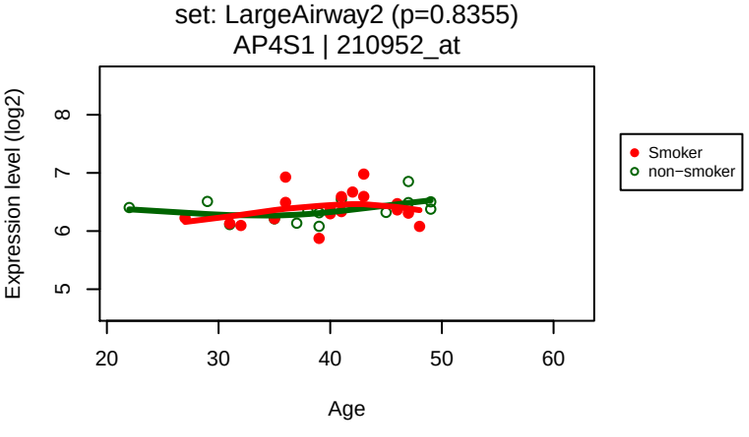

Gene expression ratio (smokers/nonsmokers)
gene: AP4S1 | 210952_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0756
p=0.1245
p=0.0317
p=0.7849
p=0.8355
p=0.3061
p=0.4964
p=0.2949
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

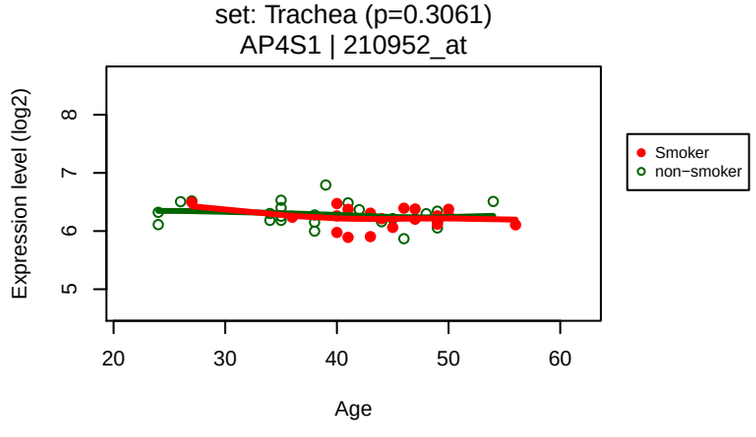

Gene expression ratio in Males
gene: AP4S1 | 210952_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5714
p=0.6508
p=0.8354
p=0.4411
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AP4S1 | 210952_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2626
p=0.6238
p=0.2454
p=0.4957
SmallAirway2
LargeAirway2
Trachea
Macrophage2

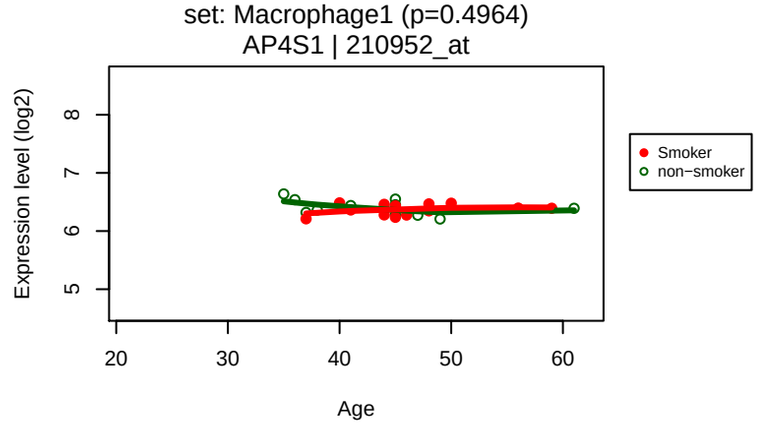

set: SmallAirway1 (p=0.0756)
AP4S1 | 210952_at
Expression level (log2)
Age
Smoker
non−smoker

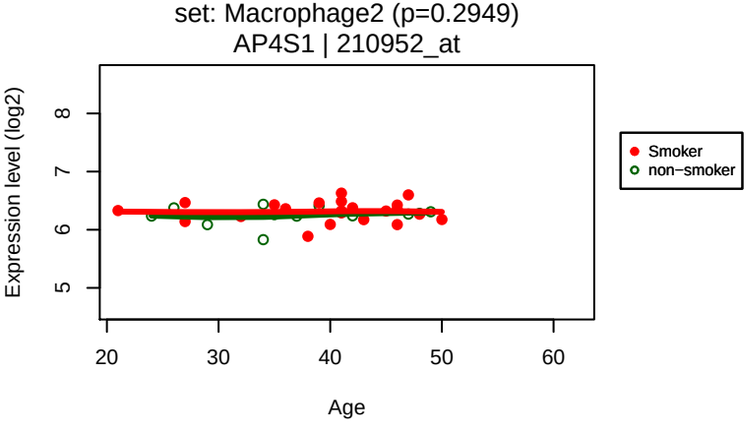

set: SmallAirway2 (p=0.1245)
AP4S1 | 210952_at
Expression level (log2)
Age
Smoker
non−smoker

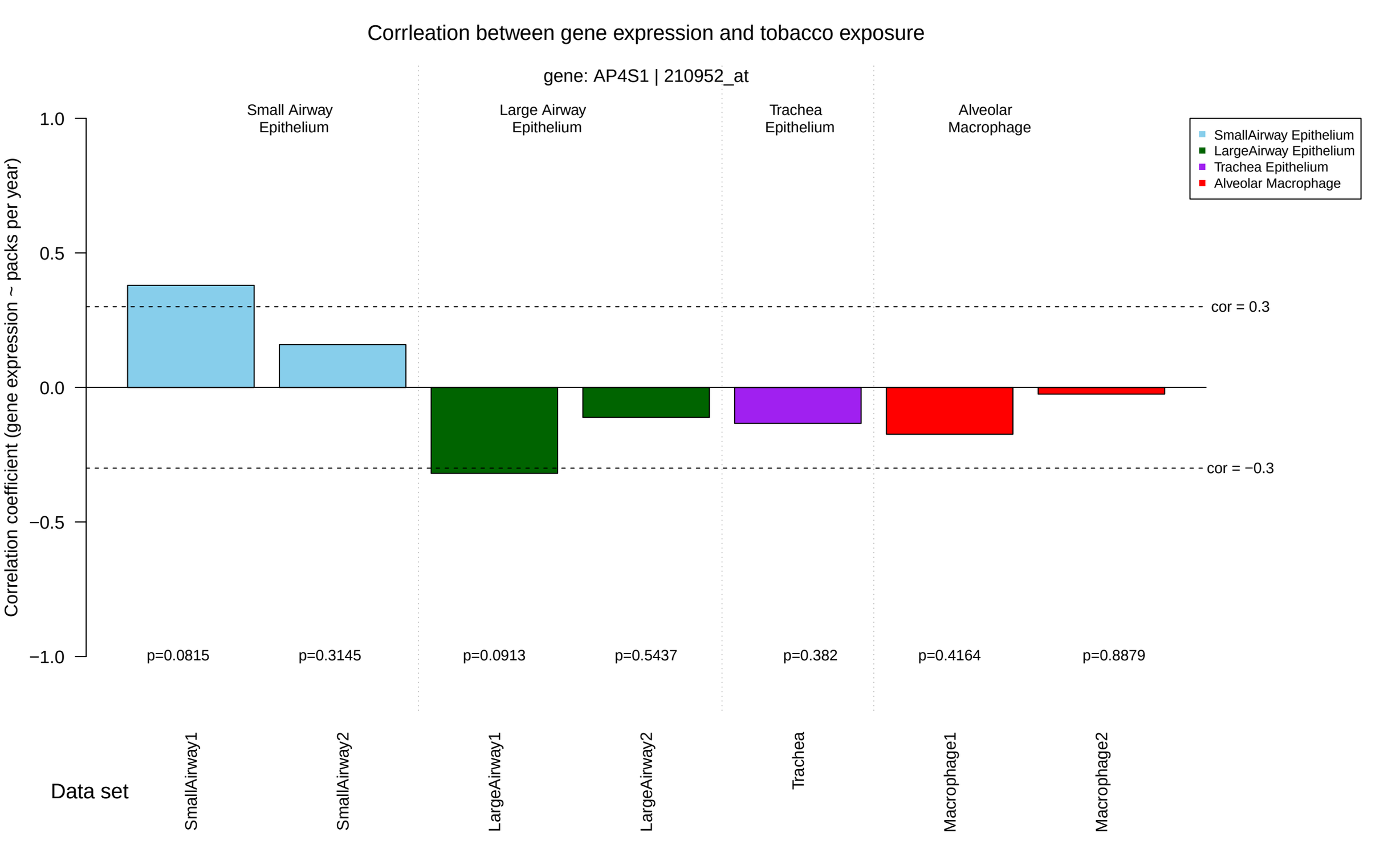

set: LargeAirway1 (p=0.7849)
AP4S1 | 210952_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8355)
AP4S1 | 210952_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3061)
AP4S1 | 210952_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4964)
AP4S1 | 210952_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2949)
AP4S1 | 210952_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AP4S1 | 210952_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0815
p=0.3145
p=0.0913
p=0.5437
p=0.382
p=0.4164
p=0.8879
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2